

Minh Pham

C/C++ Developer · Firmware Engineer

Hochiminh City, Vietnam

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[LinkedIn](#) | [Website](#) | [GitHub](#)



Objective

Looking for a career as C/C++ Developer, Firmware Engineer.

Education

California Polytechnic University, Pomona

BS – Electrical & Computer Engineering

March 2012

University of North Florida

BS – Electrical & Computer Engineering

2014 – 2015

Certifications

- IEEE Wireless Communications Professional — certification earned in Nov 2018
- HackerRank Problem Solving (Basic), LeetCode

Domain-Specific Background

- **Automotive Software Development:** familiar with AUTOSAR Classic and AUTOSAR Adaptive.
- **Android Platform:** Android Binder as IPC/RPC method; Looper / MsgQueue / Handler for threading and communication.
- **Advanced knowledge of 5G:** focus on MAC protocols.
- **Embedded systems:** understanding of **SPI, I2C, UART, CAN, LIN** protocols, boot procedure, interrupts, and the **AUTOSAR** platform.
- **Software QC testing:** design and perform manual and automation tests on 5G products. Familiar with test process and the V-Model.

Software Tools

Agentic AI

C/C++ (C++17)

Google Unit Test

Qt Framework

Slint

Rust

Flutter

Flutter Engine

C#

Ranorex Test Framework

Python

Robot Framework

QXDM

Wireshark

Professional Experience

FPT Software — Danang, Vietnam

Senior Architect, Technical Lead

Jan 2024 – Present

Trained on AUTOSAR embedded know-how

Self-training from FPT's internal knowledge base, covering the following skills:

- Configure NVIC interrupt priorities and Icu/Gpt input-capture channels to keep interrupt latency within the timing budget for time-critical signals.
- Debug bus-fault and hard-fault issues: decode CFSR, BFAR, and HFSR registers and correlate against the exception stack frame to confirm the offending peripheral access.

Team management

- Perform the roles of Senior Architect and Technical Lead. Manage tickets and project progress; work as Agile master.
- Review architecture design and system design. Review functional and non-functional features for ECUs.
- Review code before submission to customers, ensuring compliance with best practice and testability.
- Proactively make technical presentations, engage in technical discussion with the customer to clarify requirements, and find the best solution.

Programming tasks

- Demo HMI development application for Vehicle Infovista (*6 months*)
 - Implement HMI features. Fluent in Dart. Implement common components; design data flow between components and backend. Implement gRPC service in Dart to handle requests to/from backend.
 - Implement Bluetooth using the BlueZ Dart package.
 - Support implementing a Wi-Fi service in C++, running as a daemon service in Linux.
 - Implement an Observer class to watch for changes in peripherals and service notifications from the Linux kernel, such as availability of media service, Wi-Fi network, and connected devices.
 - Use the flutter-embedded-linux engine to support multi-window front-end applications.
 - Write CMake to build multiple binaries for different applications (Media, Wi-Fi, Bluetooth). Create custom CMake functions to better manage build scripts and dependencies for each application/module.
 - Write C++ backend logic to handle requests when a user clicks a front-end component and opens a new application running in a different process.

- Migrate backend library, removing Qt dependency (*6 months*)
 - Migrate Qt-dependent code to non-Qt code using C++ and Rust.
 - Implement a custom protocol for CNC machines: ensure full duplex, multiple access, encryption, and message integrity.
 - Use the Asio library to replace QNetwork; KDBinding to replace Qt signal/slot; SQLiteCpp to replace Qt SQL; `std::variant` and other C++20 containers to replace Qt containers and Qt-defined data types.
 - Implement custom helper functions for commonly used logic such as parsing message headers and converting/checking data types.
 - Use variadic template functions and spdlog to replace Qt logging.
 - Implement gTests for all production code, ensuring above 84% coverage for C0, C1, and C2. Use parameterized tests to cover a wide range of input data for the same test scenarios.
- Implement Slint components (*3 months*)
 - Define Slint custom buttons and custom layouts. Adhere to declarative coding style. Follow strict customer coding rules to create well-maintainable components.
 - Implement Rust logic to drive the Slint components.

TMA Solutions — Hochiminh City, Vietnam

Senior Software Engineer

Oct 2023 – Dec 2024

Involved in software development for the Distributed Unit (DU).

Programming tasks

- Resolve memory leak issues.
- Implement RLC for both AM/UM modes and MAC counters according to 3GPP and O-RAN requirements.
- Implement abnormal behavior handling during cell delete, involving the NG and F1AP interfaces.
- Implement UL power control algorithm and propose a UL scheduling algorithm based on both the Power Headroom Report and UL power control.
- Implement automation tests using the Robot Framework.

Team coworking

- Perform code review for team members before merge, ensuring clean-code practice.
- Provide 5G and *clean code* training.

Rikkeisoft — Hanoi, Vietnam

Junior Software Engineer at AllianceBernstein

July 2022 – Oct 2023

Involved in software maintenance and new feature delivery for AllianceBernstein.

Programming tasks

- Designed and implemented a new daemon to process account balances at end of day based on the status of data in SQL Server.

- Designed and implemented a new service handler to process new transaction requests sent to a SOAP server. Integrated the service handlers with the gSOAP library.
- Programmed automation code using C# and the Ranorex framework to read test data from Excel files, invoke tested programs, and output test results.
- Implemented stored procedures and SQL query C++ classes to send requests to the SQL database.

Software maintenance tasks

- Monitored servers running Control-M jobs for failed jobs; debugged errors by analyzing logs and code to find root causes.

LGE — Hanoi, Vietnam

Embedded System Engineer – Senior Research Engineer

Sep 2020 – July 2022

Involved in design and implementation of the route-service module in Telematics products.

Programming tasks

- Developed middleware modules (Routing Service, HMI Service, SMS Service) for embedded systems.
- Implemented interfaces and APIs to pass requests and retrieve data asynchronously via Android Binders.
- Coded in compliance with AUTOSAR platform requirements.

Research tasks

- Researched and utilized the Telux SDK and Qualcomm IPA driver to support routing service (layer 3) and network connection.
- Diagnosed issues and bugs by investigating QXDM logs, proprietary logs, and interworking between various modules.

Vinsmart — Hanoi, Vietnam

QA Engineer

Sep 2019 – Sep 2020

- Involved in quality assurance for Layer 2 operation. Tasks included designing and running test cases, debugging issues, and communicating with 3rd-party vendors to address them.
- Improved communication skills, team training, and technical writing.

Programming tasks

- Developed a DCI decoding tool and a TBS calculation tool with Java and Python to support testing tasks. Wrote C code to simulate PUCCH resources used during common procedures.

Team training tasks

- Presented 3GPP specifications to teammates on topics such as DL-CCH, UL-CCH, DMRS, HARQ, MCS, and TBS determination.

Hobbies and Other Activities

- Writing technical blogs at [Technical Blogs – My Sky](#)

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